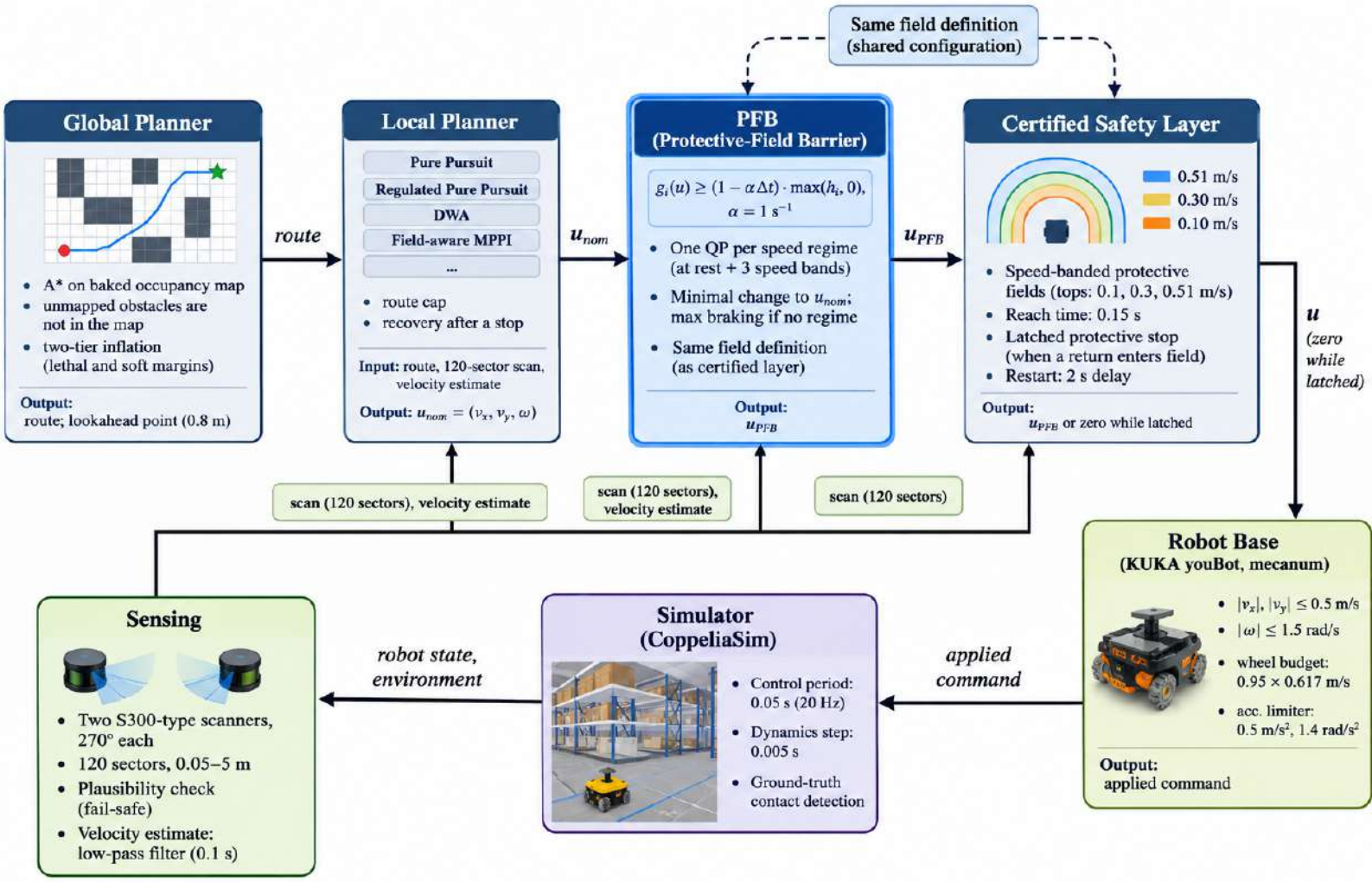

Same field definition
(shared configuration)
Global Planner
A* on baked occupancy map
unmapped obstacles are not in the map
two-tier inflation (lethal and soft margins)
Output:
route; lookahead point (0.8 m)
route
Local Planner
Pure Pursuit
Regulated Pure Pursuit
DWA
Field-aware MPPI
...
route cap
recovery after a stop
Input: route, 120-sector scan, velocity estimate
Output: $u_{nom} = (v_x, v_y, \omega)$
u_{nom}
PFB
(Protective-Field Barrier)
$g_i(u) \geq (1 - \alpha\Delta t) \cdot \max(h_i, 0)$,
$\alpha = 1\ s^{-1}$
One QP per speed regime (at rest + 3 speed bands)
Minimal change to u_{nom}; max braking if no regime
Same field definition (as certified layer)
Output:
u_{PFB}
u_{PFB}
Certified Safety Layer
0.51 m/s
0.30 m/s
0.10 m/s
Speed-banded protective fields (tops: 0.1, 0.3, 0.51 m/s)
Reach time: 0.15 s
Latched protective stop (when a return enters field)
Restart: 2 s delay
Output:
u_{PFB} or zero while latched
u
(zero while latched)
scan (120 sectors), velocity estimate
scan (120 sectors), velocity estimate
scan (120 sectors)
Robot Base
(KUKA youBot, mecanum)
$|v_x|, |v_y| \leq 0.5$ m/s
$|\omega| \leq 1.5$ rad/s
wheel budget: 0.95 × 0.617 m/s
acc. limiter: 0.5 m/s², 1.4 rad/s²
Output:
applied command
applied command
Simulator
(CoppeliaSim)
Control period: 0.05 s (20 Hz)
Dynamics step: 0.005 s
Ground-truth contact detection
robot state, environment
Sensing
Two S300-type scanners, 270° each
120 sectors, 0.05–5 m
Plausibility check (fail-safe)
Velocity estimate: low-pass filter (0.1 s)